KUNNETH FORMULA FOR EXT AGAINST PERFECT COMPLEXES ON BOTH FACTORS

THE STACKS PROJECT AUTHORS

Editorial note. No omission markers outside the final compatibility sentence.

Allowed standard prerequisites (editorial).

- Duals of perfect complexes. Cohomology, lemma dual-perfect-complex (cross-chapter, named). R SheafHom(K, K') = K' tensor K-dual for K perfect.
- Kunneth formula. lemma-kunneth (carded L26781). The box-tensor case reduces to the derived-pushforward Kunneth isomorphism.

Source reference. Original reference 08DQ: focused-seeds/geometry/S9-perf2-kunneth-Ext/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original equation 0FXY: sources/stacks-perfect/pinned-original-source.tex, source lines 6523–6523. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0FLQ: sources/stacks-perfect/pinned-original-source.tex, source lines 5636–5636. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *In the situation above, assume a and b are quasi-compact and quasi-separated and X and Y are tor independent over S. If K is perfect, $K' \in D_{QCoh}(\mathcal{O}_X)$, M is perfect, and $M' \in D_{QCoh}(\mathcal{O}_Y)$, then (0FXY) is an isomorphism.*

Proof. In this case we have $R\mathcal{H}om(K, K') = K' \otimes^{\mathbf{L}} K^\vee$, $R\mathcal{H}om(M, M') = M' \otimes^{\mathbf{L}} M^\vee$, and

$$R\mathcal{H}om(K \boxtimes M, K' \boxtimes M') = (K' \otimes^{\mathbf{L}} K^\vee) \boxtimes (M' \otimes^{\mathbf{L}} M^\vee)$$

See Cohomology, Lemma 08DQ and we also use that being perfect is preserved by pullback and by tensor products. Hence this case follows from Lemma 0FLQ. (We omit the verification that with these identifications we obtain the same map.) □

Date: stacks-project:perfect.tex:local-snapshot-20261012; GNU FDL 1.2 or later.